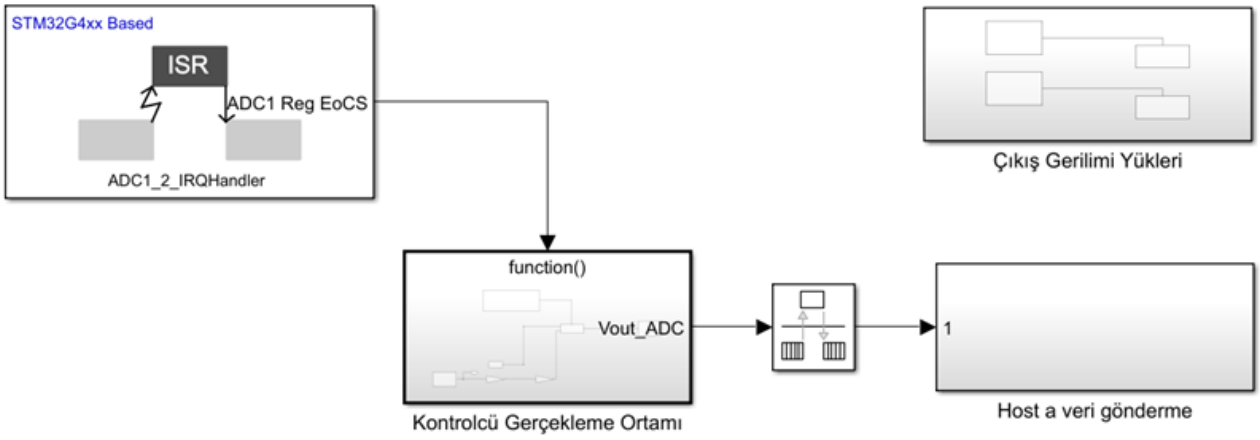
Fuzzy Logic Based Control of DC-DC Converters



Overview

This project presents the **design, simulation, and real-time implementation** of control methods for DC-DC converters, with a focus on **Fuzzy Logic Control (FLC)** applied to a buck converter.

The system is developed using MATLAB/Simulink and validated through embedded implementation.

📄 Full thesis available here:
👉 Turkish Thesis PDF

Motivation

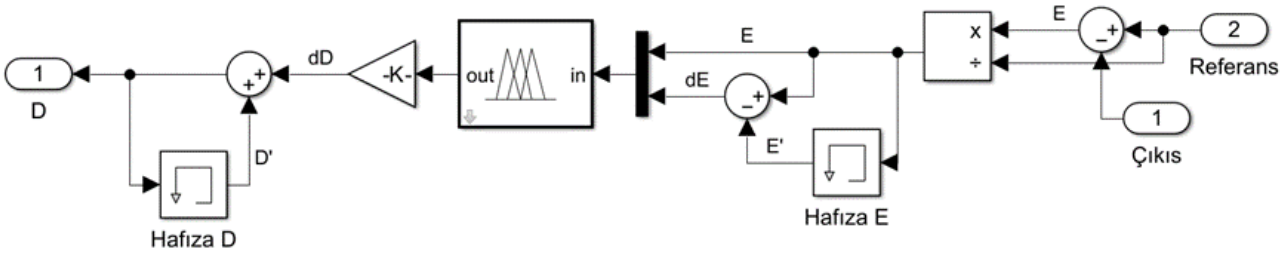
DC-DC converters operate under:

- Nonlinear dynamics
- Parameter uncertainties
- Varying load conditions

Classical controllers (PI, 3P3Z) may struggle under these conditions.

👉 This project explores **Fuzzy Logic Control** as a robust and adaptive alternative.

System Architecture

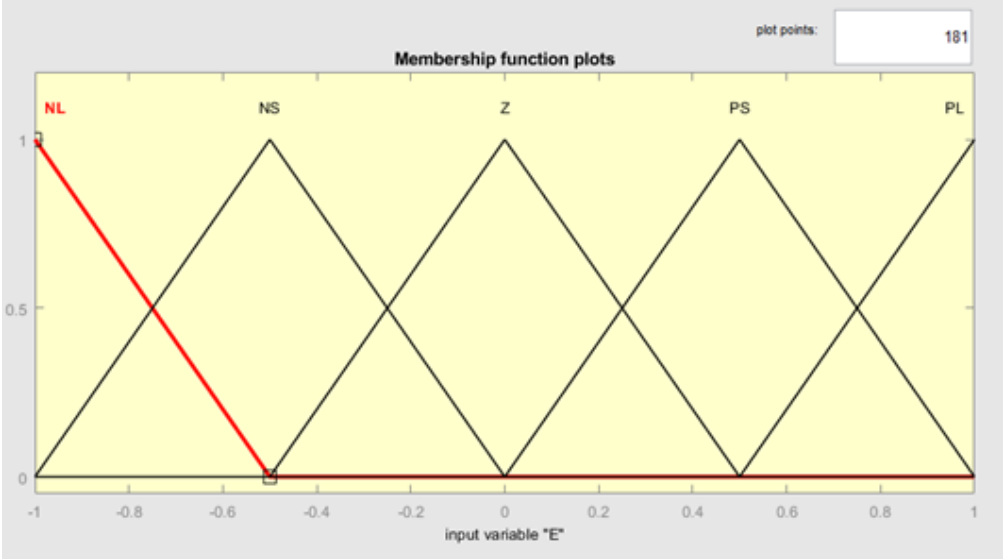


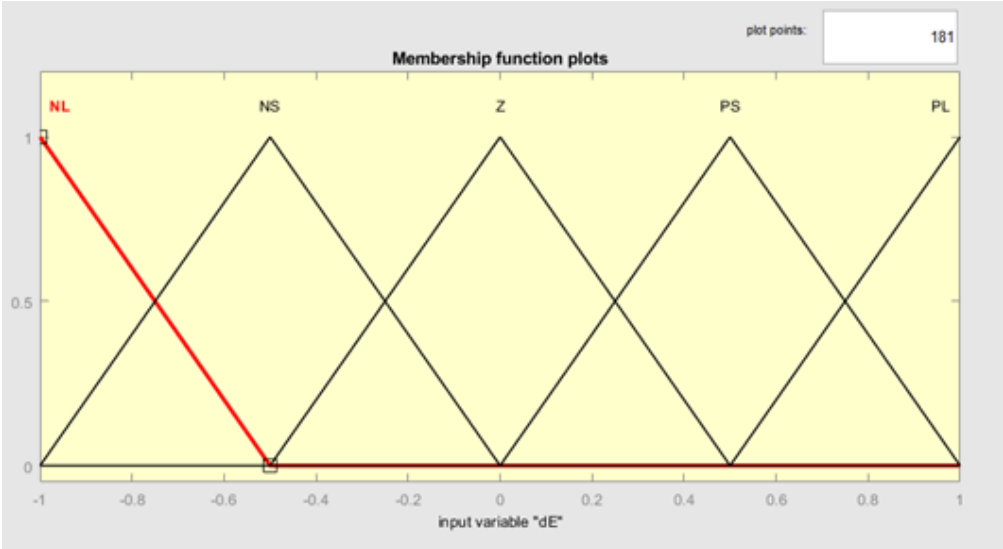
The control loop consists of:

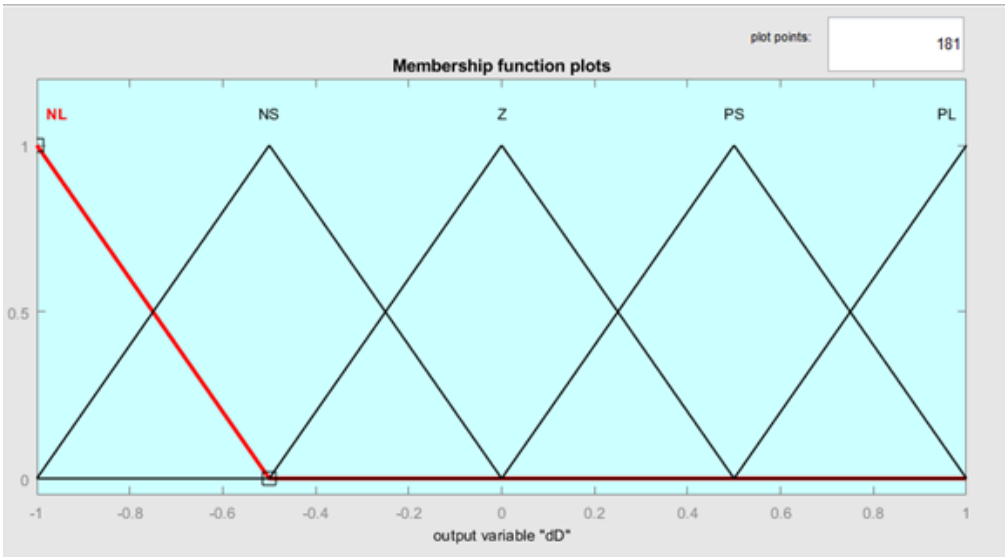
1. Output voltage measurement
2. Error and change of error computation
3. Fuzzy logic inference
4. PWM duty cycle generation

Fuzzy Logic Controller

Membership Functions







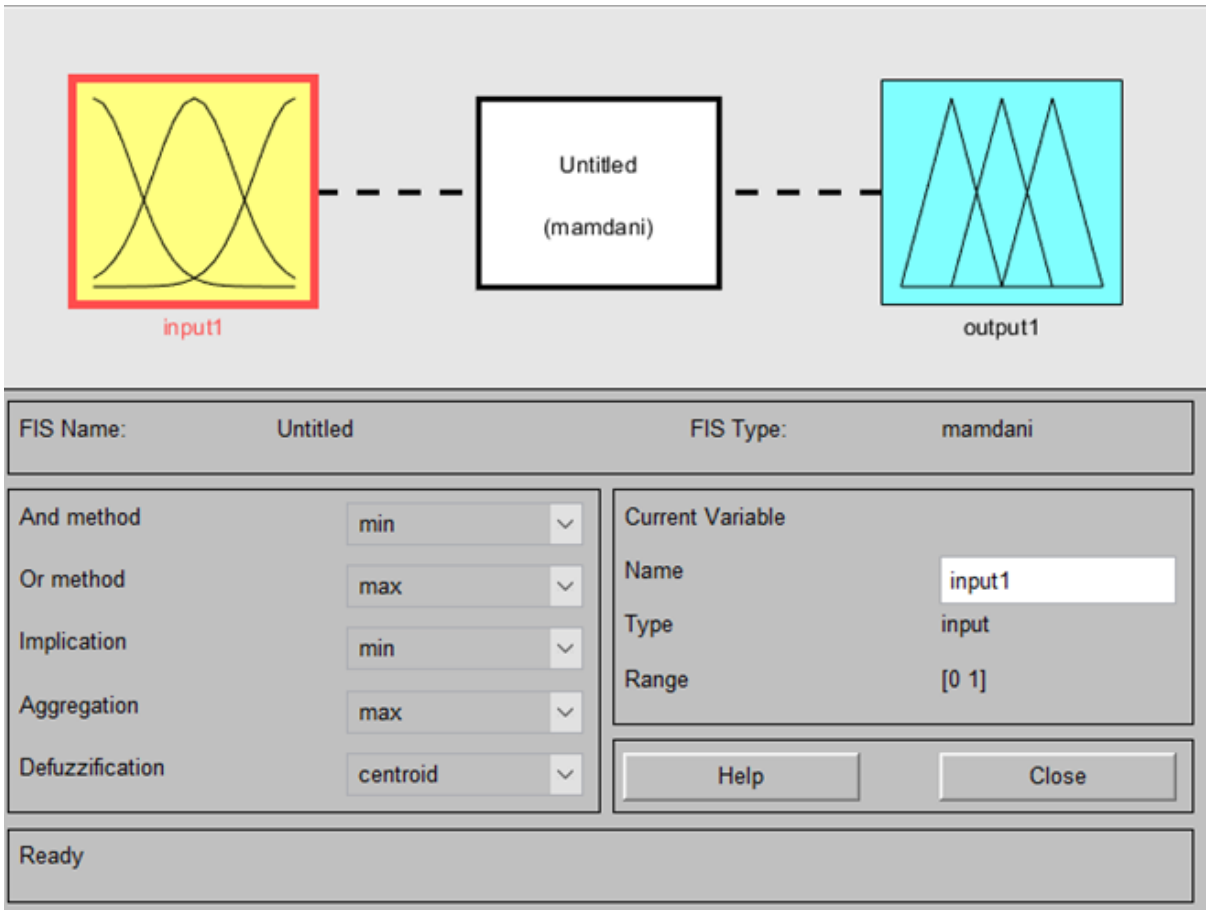
The fuzzy controller uses:

- Error (e)
- Change of Error (Δe)

Rule Base

Hata Değişimi / HATA	NL	NS	Z	PS	PL
NL	NL	NL	NL	NS	Z
NS	NL	NL	NS	Z	PS
Z	NL	NS	Z	PS	PL
PS	NS	Z	PS	PL	PL
PL	Z	PS	PL	PL	PL

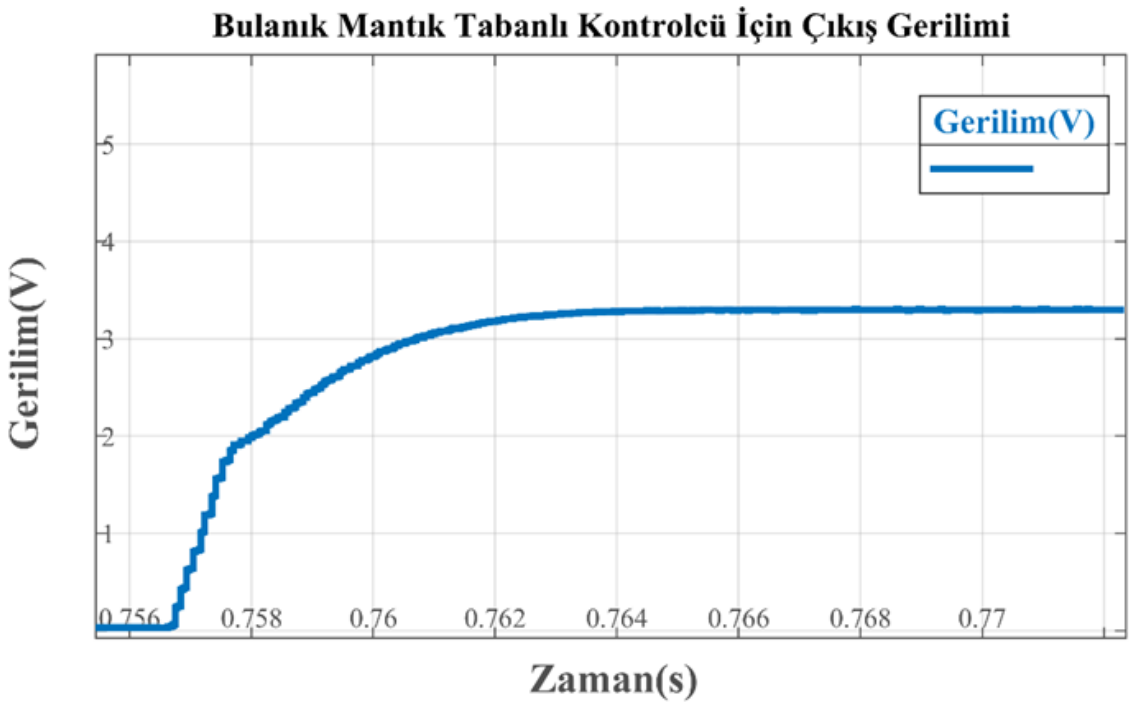
Fuzzy System Configuration



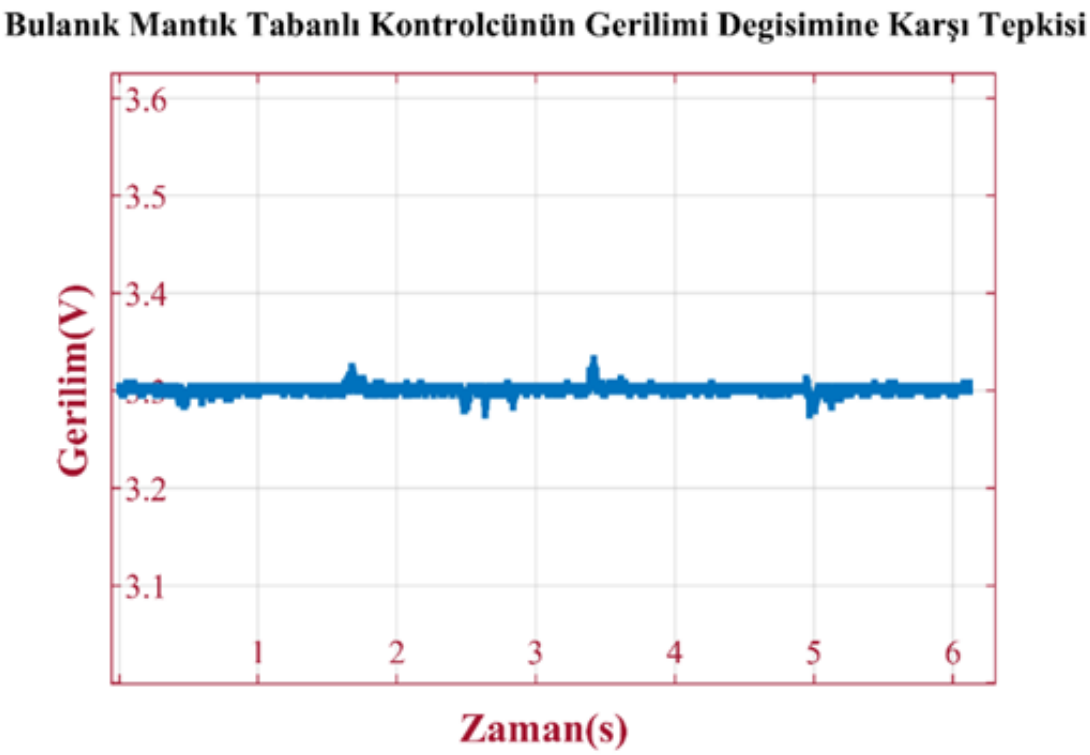
- Inference Type: Mamdani
- Defuzzification: Centroid

Simulation & Experimental Results

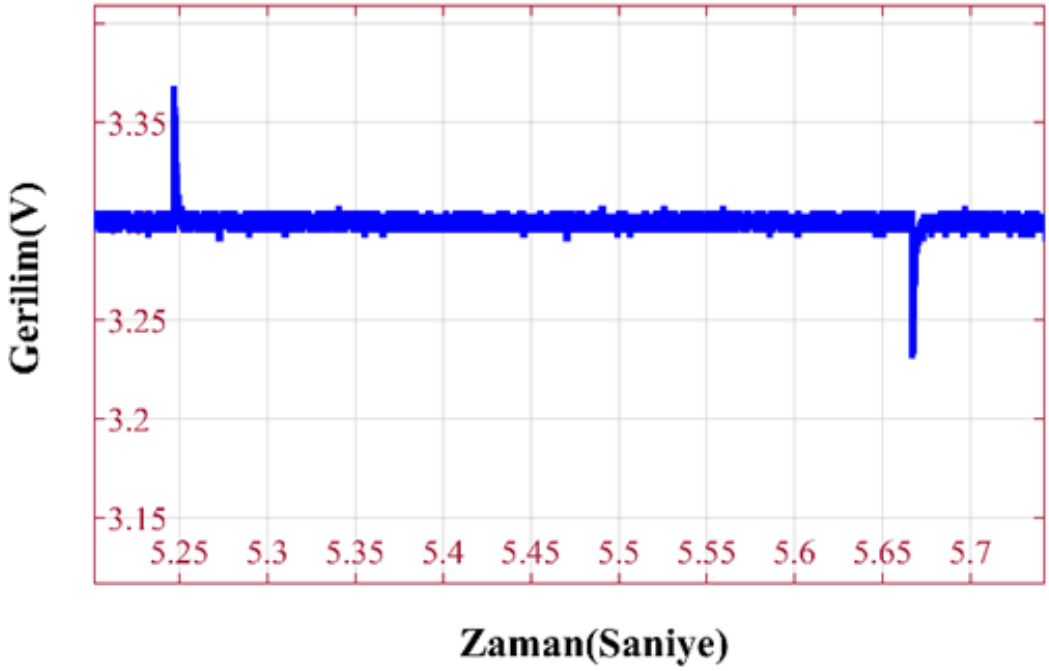
Output Voltage Response



Response to Load Changes

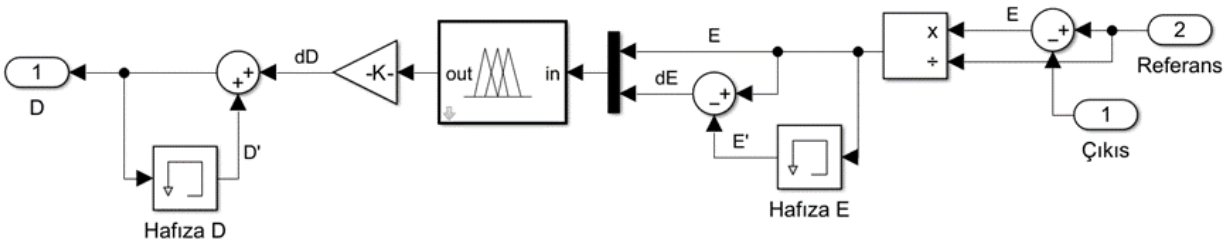


Response to Resistance Variation



Offset=0

Embedded Implementation



The controller is implemented on:

- **STM32G474E-DPOW1**

System features:

- ADC-based voltage feedback
- PWM signal generation
- Real-time control execution

Key Results

- Stable voltage regulation under disturbances
- Fast transient response
- Low overshoot
- High robustness against load and input variations

Technologies Used

- MATLAB / Simulink
- Embedded Systems (STM32)
- Power Electronics
- PWM Control
- Fuzzy Logic

Project Structure

Project Structure

```
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├── docs/
│   └── thesis_tr.pdf
├── images/
│   ├── Simulink_BlockDiagram.png
│   ├── Controller_Design.png
│   ├── Error_Membership.png
│   ├── Error_changing_membership.png
│   ├── Duty_Changing_Membership.png
│   ├── Rule_Table.png
│   ├── Fuzzy_Settings.png
│   ├── FLC_Voltage_Output.png
│   ├── FLC_Load_Changing_Performance.png
│   └── FLC_Resistor_Changing_Performance.png
└── README.md
```

Conclusion

This project demonstrates that fuzzy logic control is a strong alternative to classical controllers for DC-DC converters, particularly in nonlinear and uncertain environments.

The approach is validated through both simulation and real-time embedded implementation.

Author

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